

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electronics and Electrical Communication Engineering

2. Subject Name: Communication-I

L-T-P: 3-1-0

Credits: 4

3. Pre-requisites: -Signals and Systems(EC21208)

4. Syllabus and reference books:

Syllabus:

Representation of signals and systems in a communication system; Physical channel; Linear continuous wave (CW) modulation; Nonlinear modulation techniques; performance of analog modulation schemes; Radio transmitters and receivers; Sampling; Quantization; Digital Modulations; Carrier recovery and synchronization

Reference Books:

- 1.Communication Systems: A. Bruce Carlson, Macgraw Hill, 5th Ed, 2011
- 2.Principles of Communication Systems: Herbert Taub and Donald L. Schilling, Macgraw Hill, 2008.
3. Introduction to Communication Systems: U. Madhow, Cambridge University Press, 2014.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Representation of signals and systems in a communication system: concepts of modulation and frequency translation, lowpass and bandpass signals, concept of complex envelope, Hilbert transform and phase shifting	6
2.	Physical channel: Input output modeling, lowpass and bandpass channels, group delay and phase delay, distortion, equalization	4
3.	Linear continuous wave (CW) modulation: AM, DSB/SC, SSB, VSB, methods of generation; Demodulation techniques of CW modulation: coherent and non-coherent	4
4.	Nonlinear modulation techniques: FM and PM, narrowband FM, wideband FM, methods of generation, FM spectrum, Demodulation techniques for FM	4
5.	Performance of analog modulation schemes in AWGN: CNR, post-demodulation SNR and figure of merit for AM, DSB/SC, SSB, FM, threshold effect in FM, pre-emphasis and de-emphasis in FM.	4
6.	Radio transmitters and receivers; Noise in receivers; Noise figures;	4

	Radio link design	
7.	Sampling a signal by periodic pulse stream: spectra of ideally sampled signal, Nyquist sampling theorem, sampling of bandpass signals	4
8.	Quantization: Scalar quantization, fidelity criteria, Max-Lloyd Algorithm	4
9.	Digital Modulations: pulse amplitude modulation (PAM), pulse width modulation (PWM), pulse position modulation (PPM), quadrature amplitude modulation (QAM), continuous phase modulation (CPM), minimum shift Keying (MSK), Frequency shift keying (FSK), Phase shift keying (PSK), orthogonal shift keying (OSK): spectra, generation and demodulation schemes; inter symbol interference (ISI), Nyquist criteria, matched filter, Comparison of modulation schemes	10
10.	Carrier recovery and synchronization, phase locked loop	4
Total number of hours		48